

MOHAMMED SOHAIL

Cloud, DevOps & Full-Stack Developer | AWS • Kubernetes • Automation

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PROFESSIONAL SUMMARY

Hands-on Cloud, DevOps & Full-Stack Developer with a solid foundation in Electronics & Instrumentation Engineering, cloud infrastructure automation, container orchestration, and full-stack software architecture. Demonstrated practical experience provisioning and managing dual Kubernetes environments (self-managed kubeadm on EC2 and AWS EKS), authoring modular Terraform Infrastructure-as-Code, orchestrating automated GitOps delivery pipelines with GitHub Actions and ArgoCD, and developing production full-stack TypeScript/React and Python applications. Proven track record in diagnosing and resolving complex production incidents spanning Kubernetes CSI storage provisioners, reverse proxies, and IAM governance.

TECHNICAL SKILLS

Frontend: React 19 / 18, TypeScript, JavaScript, Vite, Tailwind CSS, HTML5, CSS3, Framer Motion, Responsive UI

Backend & APIs: Node.js, Hono, Express, Python, Django 5, tRPC, REST APIs, Zod, Gunicorn

Databases & Data: PostgreSQL, Neon PostgreSQL (Serverless), Redis, Drizzle ORM, StatefulSets Data Persistence

DevOps & Cloud: AWS (EKS, VPC, EC2, S3, RDS, ALB, CloudFront, Route 53, IAM/IRSA, CloudWatch), Kubernetes, Docker, Terraform, Helm, ArgoCD, GitHub Actions, Jenkins, Ansible, kubeadm, Calico CNI, NGINX

Systems & Observability: Linux Administration (Ubuntu), Bash Scripting, TCP/IP, OSI Model, Subnetting, NAT, Prometheus, Grafana, CloudWatch, Git

PROFESSIONAL EXPERIENCE

Visys Cloud Technologies — DevOps Engineering Intern

December 2025 – June 2026

- Automated end-to-end continuous integration and deployment pipelines using GitHub Actions, Jenkins, Docker, and Helm to accelerate release velocity.
- Provisioned and maintained resilient AWS infrastructure using Terraform IaC across multiple availability zones adhering to least-privilege security.
- Deployed, scaled, and managed containerized workloads across both managed Amazon EKS clusters and self-hosted Kubernetes (kubeadm) environments.
- Standardized Linux server configurations, user access controls, and repetitive operational maintenance routines using Ansible playbooks and Bash automation.
- Integrated multi-stage Docker builds and Helm release packaging to enforce reproducible, immutable build artifacts across staging and production.

EDUCATION

Muffakham Jah College of Engineering and Technology

2019 – 2023

Electronics & Instrumentation Engineering

FEATURED PROJECTS

AM Fruits (Fresh Flow): Live B2B Wholesale Commerce Platform

Live Production: amfruits.shop

Technologies: React, TypeScript, Hono, tRPC, Drizzle ORM, Neon PostgreSQL, Razorpay API, Docker, NGINX, Cloudflare DNS, AWS

- Built and deployed a production B2B wholesale platform (amfruits.shop) supporting role-based access control (RBAC) for wholesale buyers and platform administrators.
- Engineered an immutable snapshot order system in PostgreSQL preserving product descriptions, unit pricing, and tax rates at time of transaction for auditing.
- Implemented Razorpay payment processing featuring cryptographic server-side signature verification of transaction payloads to prevent payment tampering.
- Containerized the full application stack using Docker and deployed behind an NGINX reverse proxy with Cloudflare edge DNS and SSL/TLS termination.

FEATURED PROJECTS (CONTINUED)

SmartOrder: Cloud-Native Self-Service Restaurant Commerce System

Portfolio Project ID: 35

Technologies: TypeScript, React, Vite, Hono, tRPC, Drizzle ORM, PostgreSQL (Neon Serverless), Tailwind CSS, Git/GitHub

- Designed a full-stack, touchscreen-optimized restaurant ordering system connecting a customer self-service kiosk workflow with an administrative backoffice.
- Built a normalized relational database schema in PostgreSQL using Drizzle ORM to dynamically model multi-variant product configurations (variants, sizes, option groups, and pricing modifiers) without hard-coded frontend permutations.
- Implemented end-to-end type safety between backend and frontend via tRPC and Hono API routing, preserving product configuration states across cart and checkout.
- Implemented an order lifecycle tracking system that generates human-readable short order tokens (e.g., T 2390) for counter settlement while preserving customer data privacy.

SohailShop: Dual-Cluster Kubernetes E-Commerce Platform

github.com/sohail-24/devops-ecommerce-platform

Technologies: Kubernetes (kubeadm & AWS EKS), Terraform, Helm, ArgoCD, Docker, AWS (ALB, S3, IRSA), Django 5, PostgreSQL, Redis, NGINX

- Engineered a production-grade modular e-commerce backend and deployed it across two distinct Kubernetes environments: a self-managed kubeadm cluster on EC2 and a managed AWS EKS cluster provisioned via Terraform IaC.
- Architected a 2-repository GitOps delivery pipeline: application code changes trigger GitHub Actions to build/push immutable Docker images with Git SHA tags, updating infrastructure manifests reconciled automatically by ArgoCD.
- Packaged Kubernetes manifests into modular Helm charts with configurable CPU/memory requests/limits, ConfigMaps, Secrets, and zero-downtime rolling update probes.
- Configured stateful persistence using PostgreSQL StatefulSets, local-path storage, and AWS EBS CSI drivers; offloaded static/media assets to Amazon S3 with IAM Roles for Service Accounts (IRSA) for least-privilege authorization.

DEVOPS & CLOUD INFRASTRUCTURE HANDS-ON WORK

AWS Multi-AZ High Availability Architecture: Architected fault-tolerant multi-availability zone infrastructure eliminating single points of failure across web, application, and database tiers. Configured Application Load Balancers with path-based routing, target health checks, Auto Scaling groups, and multi-AZ database replication with automated standby failover.

Kubernetes Zero-Downtime Rolling Deployments: Containerized web services and authored declarative Kubernetes Deployment manifests configuring maxSurge and maxUnavailable rolling upgrade parameters. Implemented readiness/liveness HTTP probes and validated zero-downtime updates under load using Apache Bench with 0% dropped requests.

Modular Terraform AWS VPC & Networking Blueprint: Authored modular, reusable Terraform code provisioning a VPC with isolated public, private application, and private database subnets across 2 AZs. Implemented slash-notation CIDR partitioning (/24, /28), internet gateways, and managed NAT gateways with strict security group boundaries.

Static Cloud Distribution & Edge Caching Lab: Provisioned secure cloud storage on Amazon S3 configured with CloudFront CDN distribution delivering edge-cached assets globally (<30ms latency). Enforced strict Origin Access Control (OAC), completely blocking direct public bucket access, with ACM TLS certificates and Route 53 DNS.

Linux Administration, Diagnostic Runbooks & Scripting: Developed system diagnostics runbooks analyzing OSI Layer 2–7 transport failures, MTU packet truncation, and DNS resolution latency using dig +trace and curl. Authored Bash shell scripts automating log rotations, process inspection, and systemd unit health checks across Linux server instances.

PRODUCTION INCIDENT RESOLUTION & ROOT CAUSE ANALYSIS (RCA)

Kubernetes Storage Provisioning (CSI Driver): Diagnosed PersistentVolumeClaim stuck in Pending state on a self-managed kubeadm cluster; traced failure to CSI provisioner driver configuration and storage class bindings, successfully restoring persistent volume mounts.

WSGI Reverse-Proxy Media Routing (NGINX / Unicorn): Resolved 404/broken media assets in production Django by diagnosing Unicorn's inability to serve static/media files directly; re-architected NGINX reverse-proxy rules to route media files directly from dedicated storage volumes.

AWS S3 IAM Role Access Conflicts (IRSA): Identified HTTP 500 errors during production file uploads caused by AWS account bucket mismatches; resolved conflicts between hardcoded IAM keys and IRSA roles, realigning least-privilege IAM policies.

Containerized Redis Session Failures: Diagnosed container-level HTTP 500 crashes during session writes; traced issue to inter-container DNS resolution and corrected the Redis service hostname and network bridge configuration.